

Prediction Part II

Classification, Tree, Ensemble Learning

Classification vs. Regression

From previous lectures:

Problem type	Output Space	Example Tasks
Regression	Numeric (continuous or <i>discrete</i>)	House prices, number of accidents
Classification	Categorical	Spam/non-spam, disease/no disease

A (Sometimes) Blurred Boundary

Consider the following outcomes. Are they classification or regression?

1. Hand-written digit recognition (0 to 9).
2. Housing price.
3. COVID-19 PCR test.
4. Weather forecast: probability of precipitation.

Decision tree

From the regression lecture, you have already seen decision trees used for **numeric prediction**. Here we use the same tree idea for **classification**, then ask how to combine many trees into a stronger model.

Questions

- What is a decision tree? [Tree \(graph theory\)](#)
- When and why would you use a decision tree (instead of other models)?

CART

- CART stands for **C**lassification **A**nd **R**egression **T**ree, introduced by Breiman, Friedman, Olshen & Stone (1984).

- CART is the standard tree-building framework students will see in many data-analysis tools: recursively split the feature space, then predict with the terminal regions.
- Classification trees usually split to make labels purer; regression trees usually split to reduce within-node variation.
- [More discussion in jupyter notebook](#)

Further reading: An insightful note by Leo Breiman on "[two cultures in the use of statistical modeling to reach conclusions from data](#)".

Ensemble Learning

Ensemble learning: methods that aggregate multiple models to improve learning performance.

"Many weak models, when combined cleverly, beat a single strong model." --- The ensemble learning mantra by ChatGPT 3o

Tree-based Ensemble Learning

A single decision tree is

-  fast and interpretable
-  Sensitive and prone to overfitting

Overcome these flaws using ensemble learning:

Method	Main idea	What it mainly reduces
Boosting	Build trees sequentially , each fixing the last model's errors.	Bias
Bagging	Build trees in parallel on bootstrapped data; vote/average.	Variance
Random Forest	Bagging plus feature-level randomness at each split.	Variance (even more)